AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 1 · WORKBOOK

# Predictive and Generative AI

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

## One connected objective-and-evidence record.

Identify objectives, compute probabilities, expand the prefix and design checks before independent planning.

**Keep one habit:** Retain component contracts, count tables, source passages, unsupported claims and unmeasured outputs.

MAIN LESSON · SECTION 1

## Recall: reconnect earlier ideas

> Retrieve before opening notes.

### Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Capstone Expo. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

▼ **Earlier wording, alternatives and picture transcripts**

**Capstone Expo**

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MAIN LESSON · SECTION 12

# W1 · Identify each component objective

> A combined application can classify, retrieve and generate.

## Your evidence record

- Marker prefix The marker is has counts red6,blue3,green1,total10; probabilities0.6,0.3,0.1. Expanded prefix The marker is red gives end0.8,today0.2 with different alternatives. Door prefix has open4,closed1 while current sensor reportsclosed. Evaporation note N supports liquid-to-vapour change but not permanent disappearance. Keep these examples separate.

- Classify these components by **target**: label a message request/report; continue a prefix with a sequence; retrieve a stored sensor record. Explain why the application may combine them.

**My reasoning and evidence**

MAIN LESSON · SECTION 13

## W2 · Compute the marker distribution

> Retain the conditional prefix and count denominator.

### Your evidence record

- Marker prefix The marker is has counts red6,blue3,green1,total10; probabilities0.6,0.3,0.1. Expanded prefix The marker is red gives end0.8,today0.2 with different alternatives. Door prefix has open4,closed1 while current sensor reportsclosed. Evaporation note N supports liquid-to-vapour change but not permanent disappearance. Keep these examples separate.

- For The marker is, calculate the three empirical probabilities, their sum and their meaning.

**My reasoning and evidence**

MAIN LESSON · SECTION 14

## W3 · Follow the expanded prefix

> Different alternatives require their own conditional table.

### Your evidence record

- Marker prefix The marker is has counts red6,blue3,green1,total10; probabilities0.6,0.3,0.1. Expanded prefix The marker is red gives end0.8,today0.2 with different alternatives. Door prefix has open4,closed1 while current sensor reportsclosed. Evaporation note N supports liquid-to-vapour change but not permanent disappearance. Keep these examples separate.

- After red is selected, the expanded-prefix table gives end 0.8 and today 0.2. Explain why these values cannot be substituted for the red/blue/green probabilities.

**My reasoning and evidence**

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MAIN LESSON · SECTION 16

## W5 · Design useful evidence and permission checks

> A fluent status sentence is not a sensor observation.

## Your evidence record

- Marker prefix The marker is has counts red6,blue3,green1,total10; probabilities0.6,0.3,0.1. Expanded prefix The marker is red gives end0.8,today0.2 with different alternatives. Door prefix has open4,closed1 while current sensor reportsclosed. Evaporation note N supports liquid-to-vapour change but not permanent disappearance. Keep these examples separate.

- A system reports current room status using retrieved measurements and generated sentences. Name checks for **evidence**, response quality and permission.

**My reasoning and evidence**

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MAIN LESSON · SECTION 4

# W4 · Audit a new study workflow independently

> Keep one first attempt before hints or feedback.

## Read and explain independently

Identify the three mechanisms and specify how you would check each. No actual Moon passage, classifier output, poem or reference difficulty labels are supplied for this planning question. Do not invent those results or claim measured correctness. The earlier signal-count exercise remains additional quantitative practice, not this selected assessment.

A system retrieves an approved Moon note, uses a trained classifier to predict reading difficulty, then writes a poem. Identify each **mechanism** and the **evidence** needed to check its **output**.

**My independent explanation, trace and limits**

## Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

**Independent case: Changed independent case**

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MAIN LESSON · SECTION 9

# Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

## Your evidence record

Separate Study Buddy tasks into classify, calculate, retrieve, and generate stages.

The project change and **evidence** I will keep

**Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

## Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

### Your evidence record

| Evidence | My record |
|---|---|
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

# Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

## Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

## Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

### Use this when helpful

Use W2: Compute the distribution for Generative and predictive objectives. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

## Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

### Use this when helpful

Use the worked case for Generative and predictive objectives. Proposed explanation: “Generative models do not predict”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

## Reference · Original learning target

> Use this target to inspect an explanation.

### Use this when helpful

A study tool finds a paragraph, labels its difficulty and invents practice questions. Identify the three operations and one check for each.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

## Extra practice · A complete case

> Try the question before checking the worked explanation.

### Use this when helpful

A tool labels a question biology, writes a new question and computes 7 times 8.

Identify each operation.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

## Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

### Use this when helpful

A system labels a message as spam. Another writes a new reply. Which task predicts a category and which generates content? Can the new reply contain a false fact?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

**Week 1 Test and reflect**

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

## Extra practice · Design a revealing test

> Predict before tracing or testing.

### Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

▼ Reference · Original target response space

REFERENCE · SECTION 11

# Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

## Use this when helpful

A study tool finds a paragraph, labels its difficulty and invents practice questions. Identify the three operations and one check for each.

**My explanation after practice**

Record support used: none, hint or teacher explanation.

▼ Additional practice · Analyse the signal corpus

REFERENCE · SECTION 15

# Additional practice · Analyse the signal corpus

> This retained quantitative task is separate from the selected Moon-workflow assessment.

## Earlier corpus task and saved answer

- For prefix The signal is, counts are high 2, low 5, unknown 3. Compute all probabilities and the most frequent continuation. A current measurement reports high.

  State what the most frequent continuation does and does not establish.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

# Extra practice · Use the worked picture

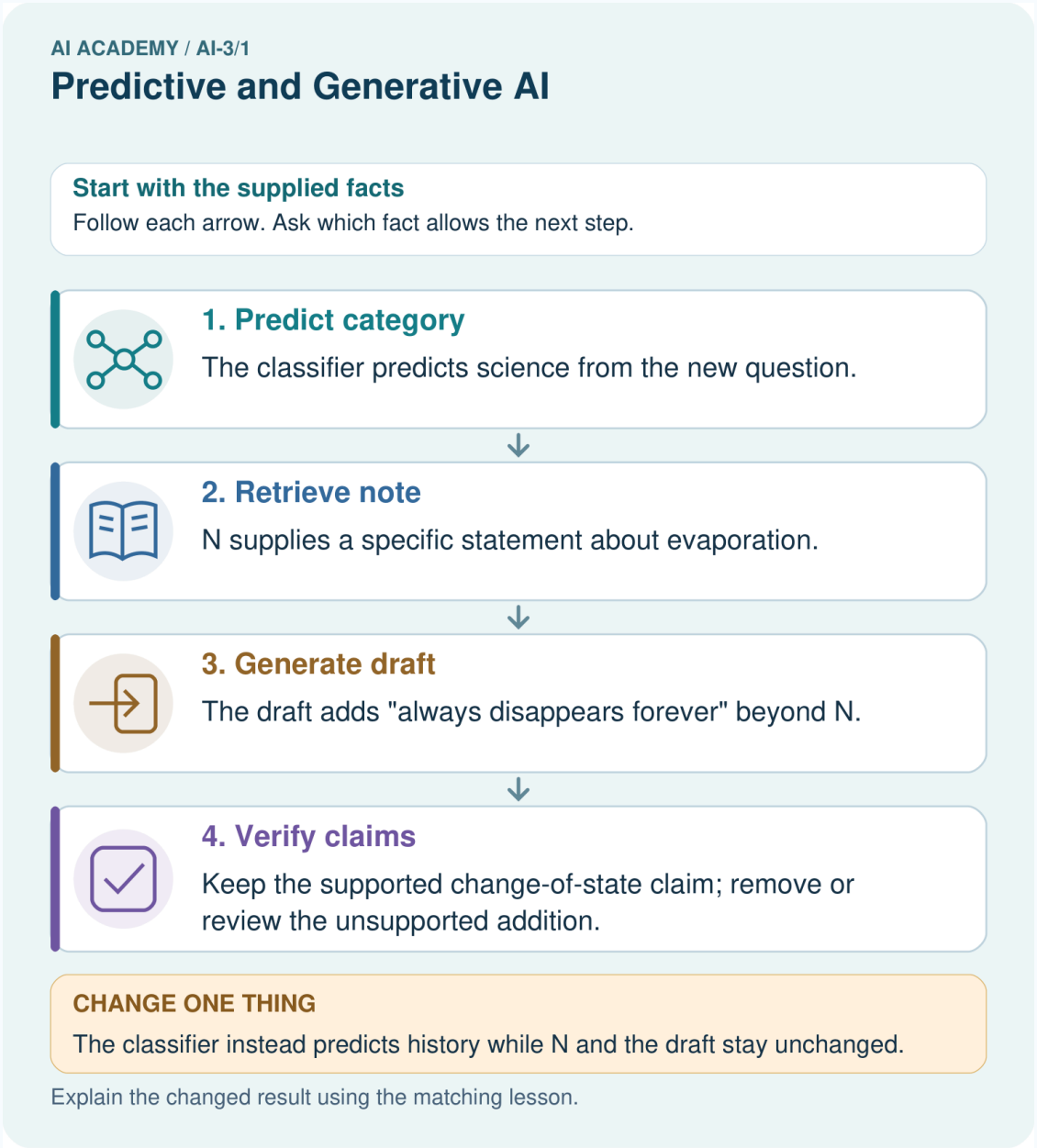
> Keep the explanation and the support you used.

## Use this when helpful

A fictional learning tool first classifies a question, then drafts an answer from a supplied note.

A fitted classifier labels a question science; its **training** used labelled questions.

Approved note N: evaporation changes liquid water into water vapour. A generator drafts: “Water evaporates into vapour and always disappears forever.”




Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the first unsupported claim and explain what source would be needed before asserting it.

> **My picture-based explanation and changed-case prediction**

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Verified Study Buddy, separate routing **prediction**s, source retrieval and generated explanations in its **evidence** record.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

# Reference · Project portfolio record

> Extend your main project response when helpful.

## Use this when helpful

Project: Verified Study Buddy

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For the Verified Study Buddy, separate routing **prediction**s, source retrieval and generated explanations in its **evidence** record.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |

| Evidence to keep | My record |
| --- | --- |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

# Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

## First supported question

- A fictional learning tool first classifies a question, then drafts an answer from a supplied note.
- Which labelled step matches this detail? Keep the supported change-of-state claim; remove or review the unsupported addition.
- 1. Predict category

- 2. Verify claims
- 3. Retrieve note
- 4. Generate draft
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Verify claims Keep the supported change-of-state claim; remove or review the unsupported addition. Classification predicts a category under a trained mapping.

Generation produces a sequence of words under a language objective. Supplying a note can support factual grounding, but the draft still requires claim-level checking.

The category being correct does not prove every sentence in the answer is correct.

## Changed-case question

- Changed-case practice: The classifier instead predicts history while N and the draft stay unchanged. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.

- 3. The category **prediction** changes, but the same unsupported draft claim still needs checking.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The category **prediction** changes, but the same unsupported draft claim still needs checking. Stages can fail independently; one result does not validate another.

## Repair and save your work

Review Capstone Expo (ai-2/36). Goal: Defend an application using reproducible **evidence**. Short practice: Give the original winning ID, its score, and counts of eligible, excluded and invalid rows..

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Repair a missing building block**

**Keep your completed work**

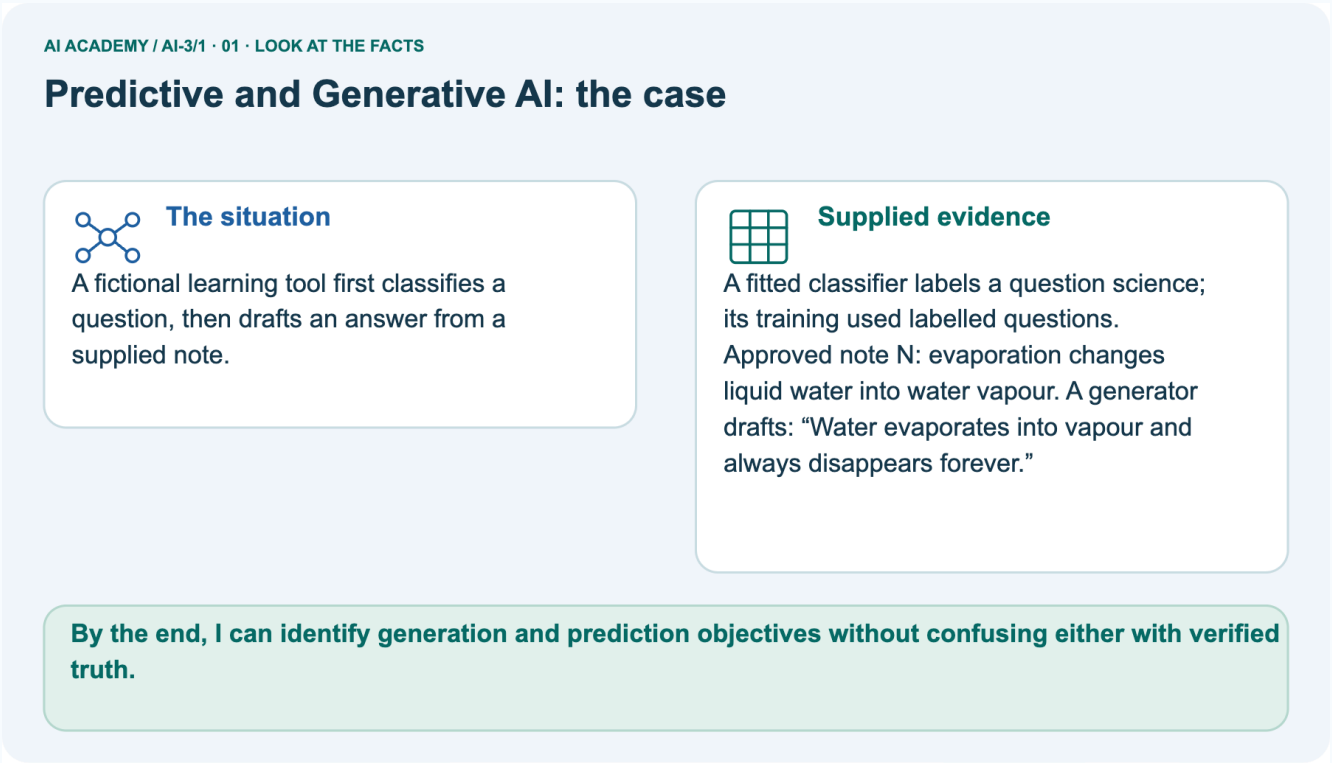
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

## Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

# 01. Predictive and Generative AI: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

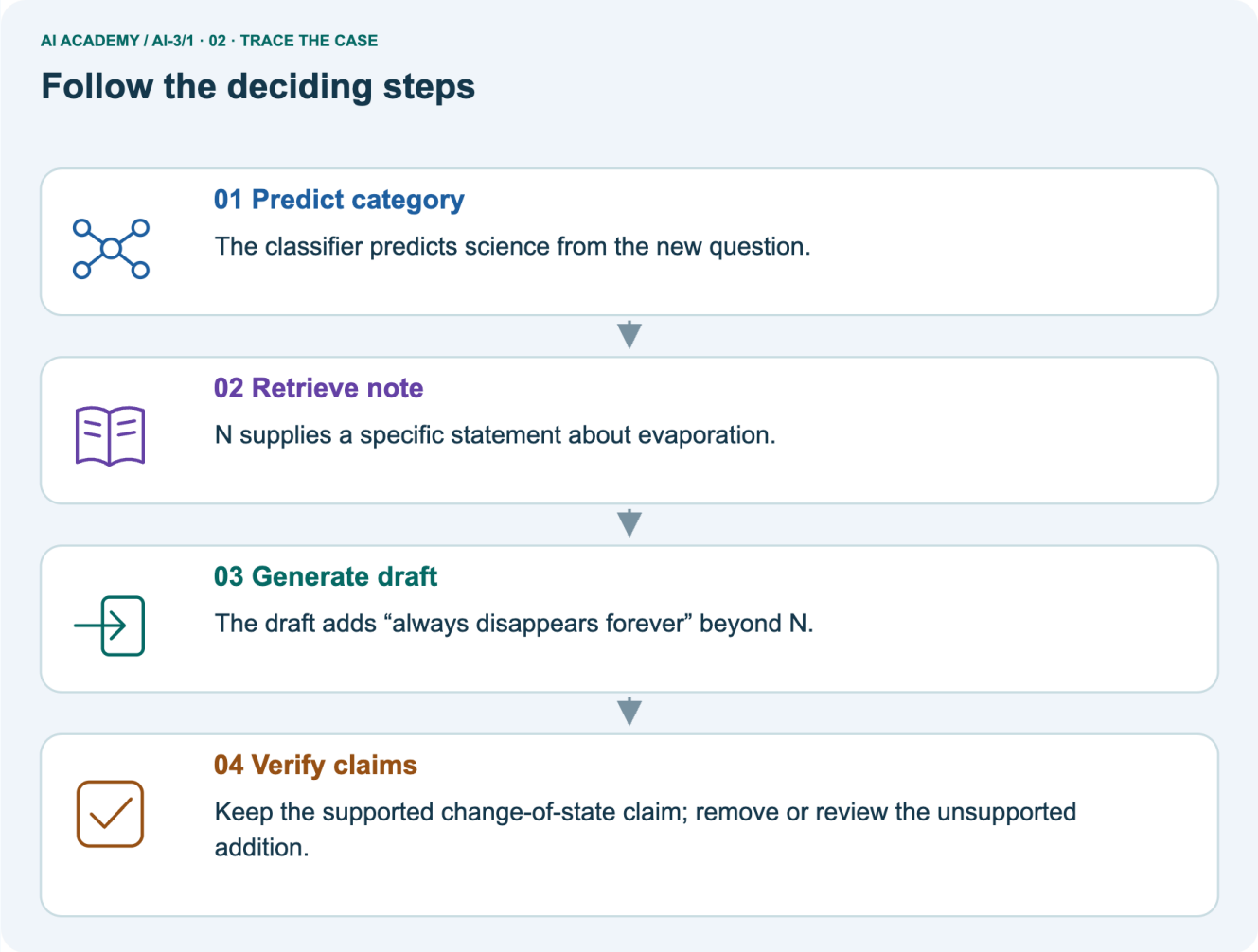
A fictional learning tool first classifies a question, then drafts an answer from a supplied note. Supplied facts: A fitted classifier labels a question science; its **training** used labelled questions.

Approved note N: evaporation changes liquid water into water vapour. A generator drafts: “Water evaporates into vapour and always disappears forever.”

Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps



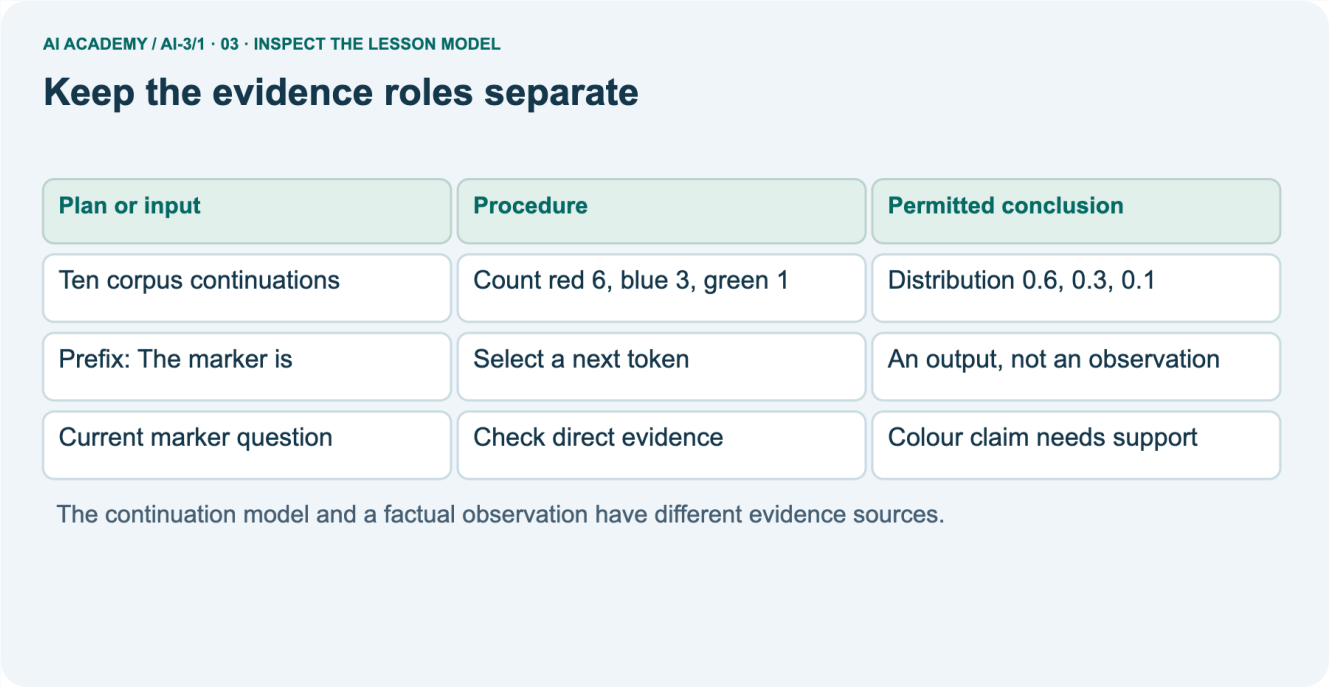

Follow the labelled picture with the matching explanation. Open the vector diagram

Predict category: The classifier predicts science from the new question. → Retrieve note: N supplies a specific statement about evaporation. → Generate draft: The draft adds “always disappears forever” beyond N. → Verify claims: Keep the supported change-of-state claim; remove or review the unsupported addition.

Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

## 03. Keep the evidence roles separate




Follow the labelled picture with the matching explanation. Open the vector diagram

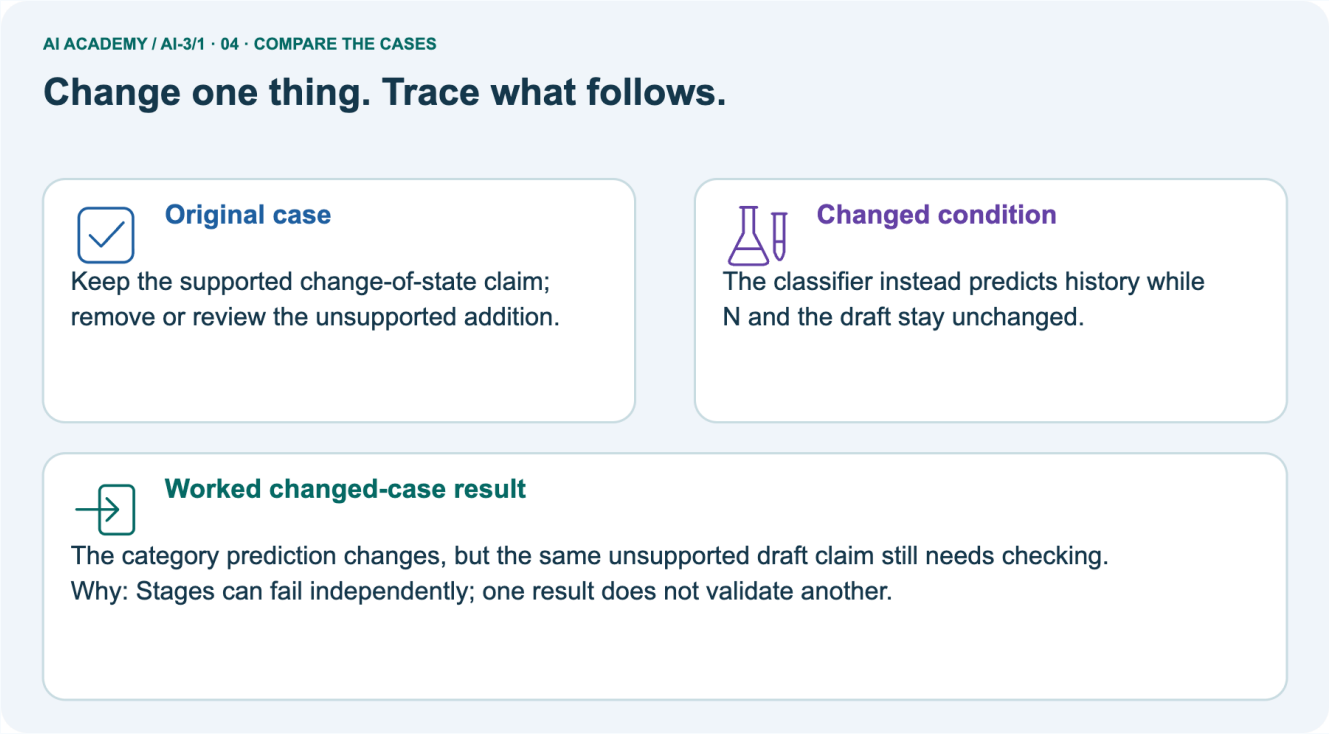
Keep the **evidence** roles separate. Ten corpus continuations | Count red 6, blue 3, green 1 | Distribution 0.6, 0.3, 0.1. Prefix: The marker is | Select a next token | An **output**, not an **observation**.

Current marker question | Check direct **evidence** | Colour claim needs support

Read the labels and values. Explain the relationship the model shows.

### My picture explanation, deciding evidence and one limit

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

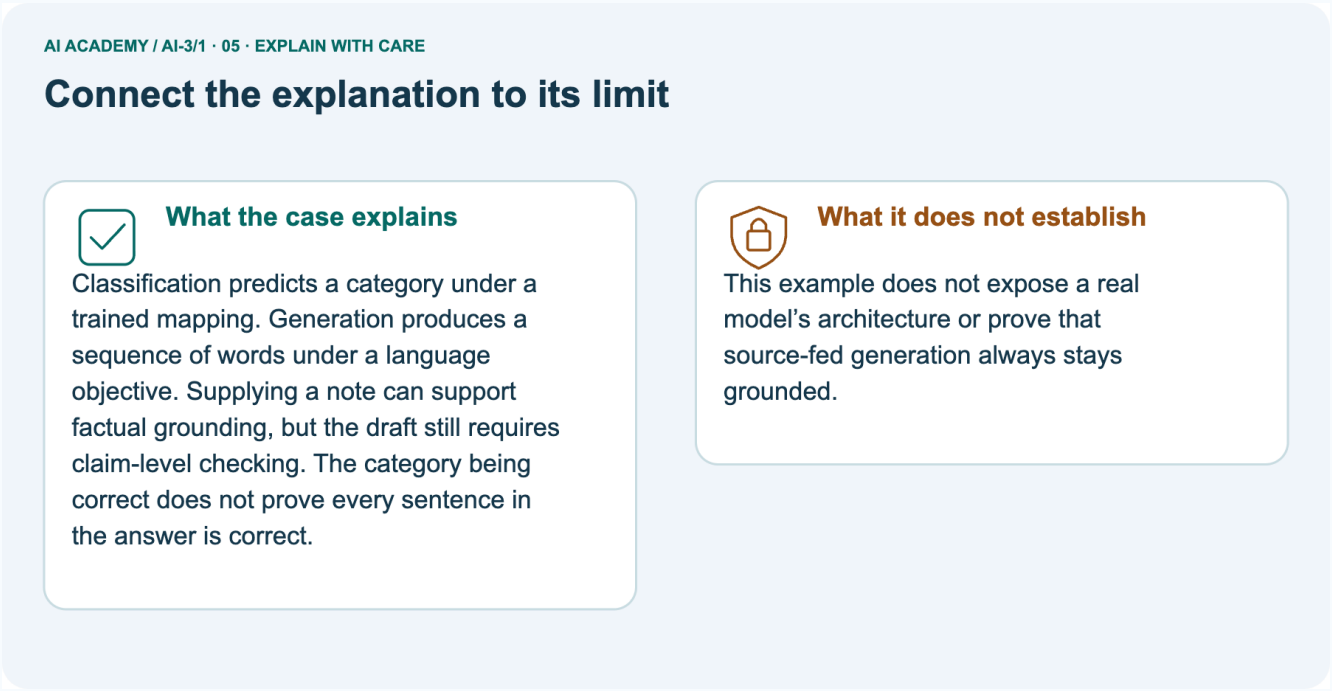
Original case: Keep the supported change-of-state claim; remove or review the unsupported addition. Changed condition: The classifier instead predicts history while N and the draft stay unchanged.

Expected result: The category **prediction** changes, but the same unsupported draft claim still needs checking. Reason: Stages can fail independently; one result does not validate another.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### My picture explanation, deciding evidence and one limit

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

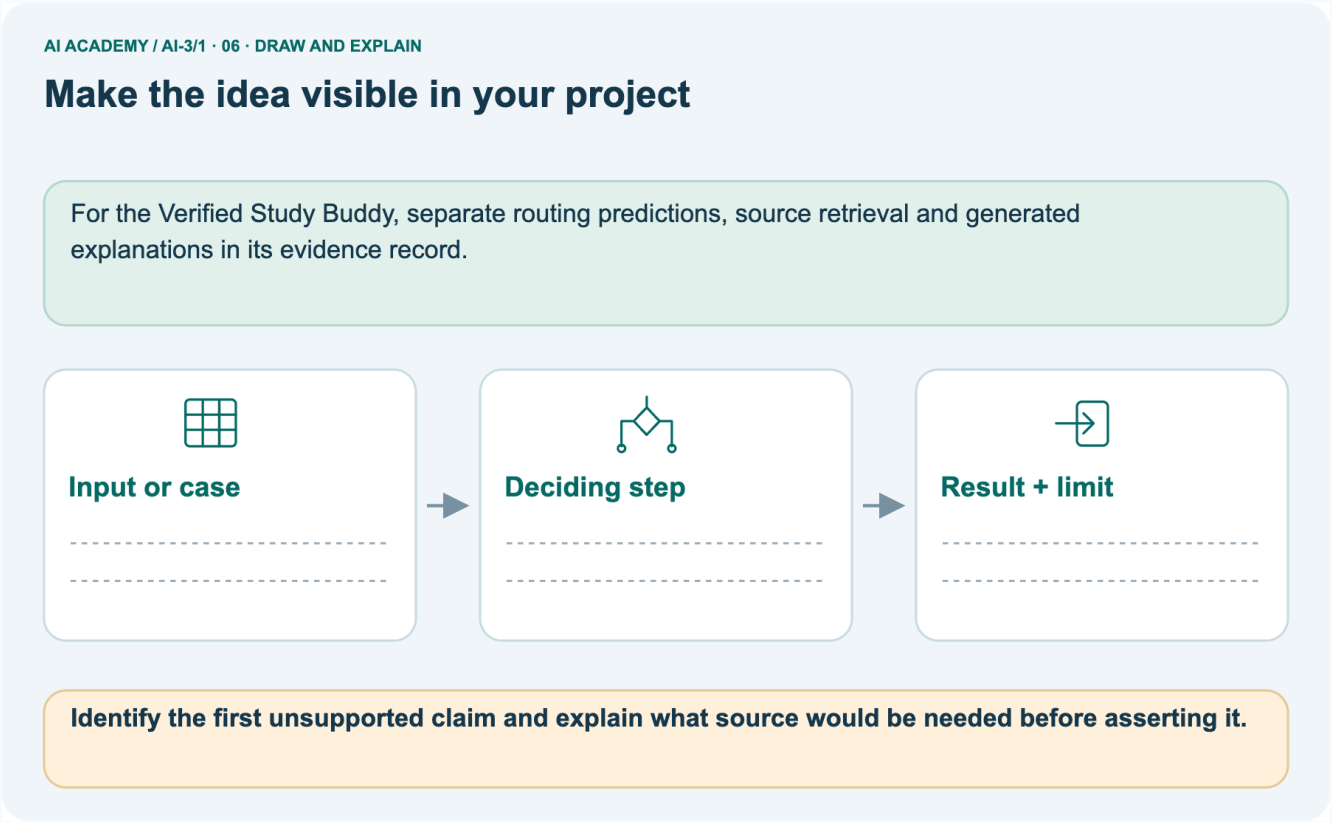
Classification predicts a category under a trained mapping. Generation produces a sequence of words under a language objective. Supplying a note can support factual grounding, but the draft still requires claim-level checking.

The category being correct does not prove every sentence in the answer is correct. Limit: This example does not expose a real model’s architecture or prove that source-fed generation always stays grounded.

Explain the case, then point to a limit on the claim.

### My picture explanation, deciding evidence and one limit

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Verified Study Buddy, separate routing **prediction**s, source retrieval and generated explanations in its **evidence** record. Draw your **input**, deciding step, result and limit.

Supported question: Identify the first unsupported claim and explain what source would be needed before asserting it.

Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

## ▼ Earlier wording, alternatives and picture transcripts

**01. Predictive and Generative AI: the case**

**02. Follow the deciding steps**

**03. Keep the evidence roles separate**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

## ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.